

MD KHALEQUZZAMAN CHOWDHURY SAYEM

Researcher, Vision & Learning Lab (UNIST), South Korea

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Research Summary

Researcher in **3D vision and vision-language models**, focusing on geometry-grounded perception and reasoning for embodied systems. My work spans real-time two-hand manipulation modeling (**AAAI 2025**) and the development of **HandVQA (CVPR 2026)**, a **1.6M-scale** benchmark diagnosing spatial reasoning failures in foundation VLMs. Across these projects, I integrate explicit **3D geometric supervision** with multimodal Transformers to improve generalization in manipulation and language-grounded tasks.

Research Interests

Physical AI; Vision-Language-Action (VLA) Models; Geometry-Grounded World Models; Multimodal Representation Learning; 3D Human-Object Interaction; Embodied Intelligence.

Education

M.S. in Computer Science and Engineering

Ulsan National Institute of Science & Technology (UNIST)

Sep 2023 – Aug 2025

Ulsan, Korea

– GPA: 3.93/4.3 (96.30/100)

– **Master's Thesis (Accepted at CVPR 2026):** *HandVQA: Diagnosing and Improving Fine-Grained Spatial Reasoning about Hands in Vision-Language Models.*

– Advisors: Prof. Seungryul Baek (Supervisor), Prof. Binod Bhattarai (Co-supervisor)

B.S. in Industrial Engineering

Ulsan National Institute of Science & Technology (UNIST)

Sep 2019 – Aug 2023

Ulsan, Korea

Selected Research Highlights

- Co-first author, **AAAI 2025**: real-time 3D two-hand manipulation modeling (**53.5 FPS**; +27.2% H2O; +10.4% FPHA).
- First author, **CVPR 2026: HandVQA** benchmark (**1.6M** geometry-grounded questions across FreiHAND / InterHand2.6M / FPHA).
- Diagnosed systematic spatial reasoning failures of foundation VLMs (e.g., depth ambiguity, left/right confusion, relational bias) and demonstrated gains from structured 3D supervision and transfer.

Conference Publications (* equal contribution)

- **[CVPR 2026]** MD K. C. Sayem*, M. T. Chowdhury*, Y. Y. Tiruneh, M. A. Khan, M. S. Ali, B. Bhattarai, S. Baek. *HandVQA: Diagnosing and Improving Fine-Grained Spatial Reasoning about Hands in Vision-Language Models.*
Project Page: kcsayem.github.io/handvqa/
 - Proposed **HandVQA**, a **1.6M-scale** benchmark derived deterministically from **3D hand pose annotations** across FreiHAND, InterHand2.6M, and FPHA.
 - Diagnosed systematic failures of large VLMs (e.g., LLaVA, Qwen2.5VL) in fine-grained spatial reasoning, revealing depth ambiguity, left/right confusion, and relational bias.
 - Demonstrated that injecting **structured 3D supervision** improves zero-shot transfer to gesture and hand-object interaction recognition tasks.
 - Provides a scalable pathway for injecting explicit **3D geometric priors** into text-first foundation models.
- **[AAAI 2025]** E. Ismayilzada*, MD K. C. Sayem*, Y. Y. Tiruneh, M. T. Chowdhury, M. Boboev, S. Baek. *QORT-Former: Query-optimized Real-time Transformer for Understanding Two Hands Manipulating Objects.*
Project Page: kcsayem.github.io/QORT-Former/
 - Designed the core query optimization method and integrated structured contact-map features into the Transformer architecture.
 - Proposed a **query division strategy** that separates queries into left-hand, right-hand, and object groups via a dedicated query proposal module, reducing ambiguity between symmetric hand features.
 - Reduced Transformer complexity from **300 queries / 6 decoders** to **108 queries / 1 decoder**, achieving **53.5 FPS** real-time inference while improving state-of-the-art accuracy by **+27.2%** (H2O) and **+10.4%** (FPHA).

Selected Research Projects

- **4D Instruct-GS2G: Editing 4D Gaussians Using Text Instructions.**
Course Project (3D Computer Vision, UNIST); Code: [GitHub](#)
 - Implemented text-conditioned editing for dynamic 3D scenes by integrating *InstructPix2Pix* with 4D Gaussian representations.
 - Achieved globally consistent edits with fast training (30 min/scene) and high rendering speed (up to 82 FPS).
- **Worker Safety System for Industrial Environments Using Multi-Task YOLOv11.**
Hyundai Motors & Vision and Learning Lab (UNIST); Code: [GitHub](#)
 - Designed and implemented a unified multi-task perception system performing detection, instance segmentation, and human pose estimation.
 - Integrated YOLOv11 detection, segmentation, and pose heads into a single pipeline, reducing inference latency by **35%** compared to separate module execution.

Research Experience

- Vision & Learning Lab, UNIST

Jan 2023 – Present

Researcher (previously Research Assistant during M.S.)

Ulsan, Korea
- Developed **geometry-aware Transformers** for articulated 3D human–object interaction and real-time inference.
 - Conducted large-scale multimodal experiments to analyze **representation bias** and **generalization** under viewpoint changes, occlusion, and egocentric settings.
 - Designed evaluation pipelines and ablations for diagnosing spatial reasoning failure modes in foundation VLMs.
- Machine Learning & Vision Lab, UNIST

Sep 2021 – Feb 2022

Research Assistant

Ulsan, Korea
- Implemented data augmentation and training improvements for 2D human pose estimation, improving robustness on COCO-style benchmarks.

Technical Skills

- **Core:** PyTorch, CUDA, Distributed/Accelerated Training, Linux, Git, LaTeX
- **Multimodal / Foundation Models:** HuggingFace Transformers, LoRA fine-tuning, VLM benchmarking (e.g., LLaVA, Qwen-VL)
- **3D Vision:** Pose estimation, multi-view geometry, human/hand modeling, NeRF / Gaussian Splatting (projects)
- **Data & Tooling:** NumPy, Pandas, OpenCV, experiment scripting and evaluation pipelines

Teaching Experience

- **Teaching Assistant:** Introduction to AI Programming in Python (Fall 23, Spring 24, Fall 24) — Instructor: Prof. Michael Jon Burrell
- **Teaching Assistant:** Discrete Mathematics (Fall 24) — Instructor: Prof. Michael Jon Burrell
- **Teaching Assistant:** Advanced Computer Vision (Spring 24) — Instructor: Prof. Seungryul Baek

Honors & Awards

- **Graduate School Scholarship (Korean Government):** full tuition + stipend for M.S.
- **UNIST Dream Scholarship (UNIST):** full tuition + stipend for B.S.
- **AI Challengers Program (UNIST Innovative Education Center):** secured **\$5K** project funding (bandit algorithm parallelization).

REFERENCES

Professor Seungryul Baek	Supervisor	Professor Binod Bhattarai	Co-supervisor
	<i>Masters</i>		<i>Masters</i>
– Affiliation: Associate Professor, Artificial Intelligence Graduate School (AIGS), UNIST, South Korea		– Affiliation: Lecturer, University of Aberdeen, Aberdeen, UK	
– Affiliation: Associate Professor, Department of Computer Science and Engineering, UNIST, South Korea		– Affiliation: Honorary Lecturer, University College London, UK	
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